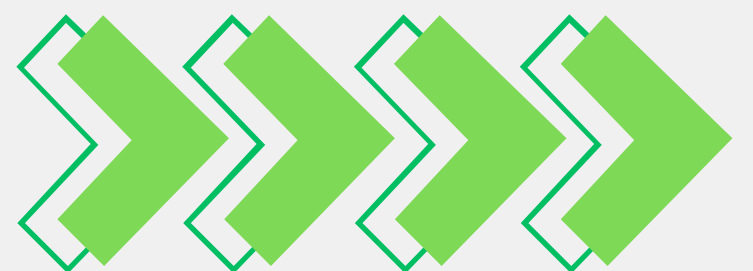


Exploring Fine-Tuning AI into a Specified Domain (Indonesian Legal) R&D

An in-depth look at leveraging
DeepSeek-R1-0528-Qwen3-8B with
Group Relative Policy Optimization
(GRPO) for structured legal
reasoning



Afza Azzindani



Bridging the Legal Accessibility Gap



- **Motivation:** Structured legal knowledge must be more accessible in Indonesia, empowering citizens and various professionals.
- **The Research Question:** How effectively can an AI be trained to not just retrieve information, but to reason through complex Indonesian legal contexts with transparency?
- **Transparent Outcomes:** I am openly exploring positive potential (democratizing knowledge for everyone) alongside challenges (identifying potential biases and ethical limitations).
- **Objective:** Develop an AI that thinks in steps, like a legal practitioner.

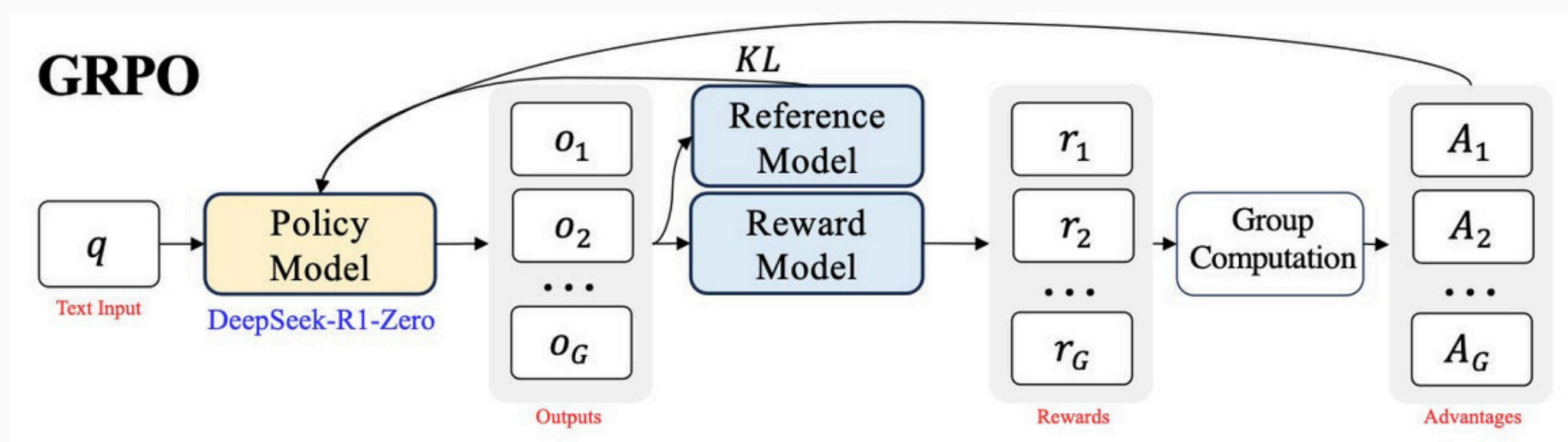


Group Relative Policy Optimization (GRPO)

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I chose GRPO for its strength in structured reasoning. While RLHF, PPO, DPO, or RLVR optimize global or pairwise rewards, GRPO groups samples by difficulty or topic, enabling step-by-step improvements much **like how students progress in mastering complex subjects**:

- **The Analogy:** Imagine the AI taking a specialized **self-test**. For every complex problem, it generates several different answers (like multiple practice attempts).
- **Relative Learning:** The model **compares each answer's score** to the average score of its own group. This forces it to learn what works best relative to its current ability.
- **Result:** The model trains itself to consistently produce outputs that are structurally superior, instilling the logical flow required for **thorough thinking** in any domain.

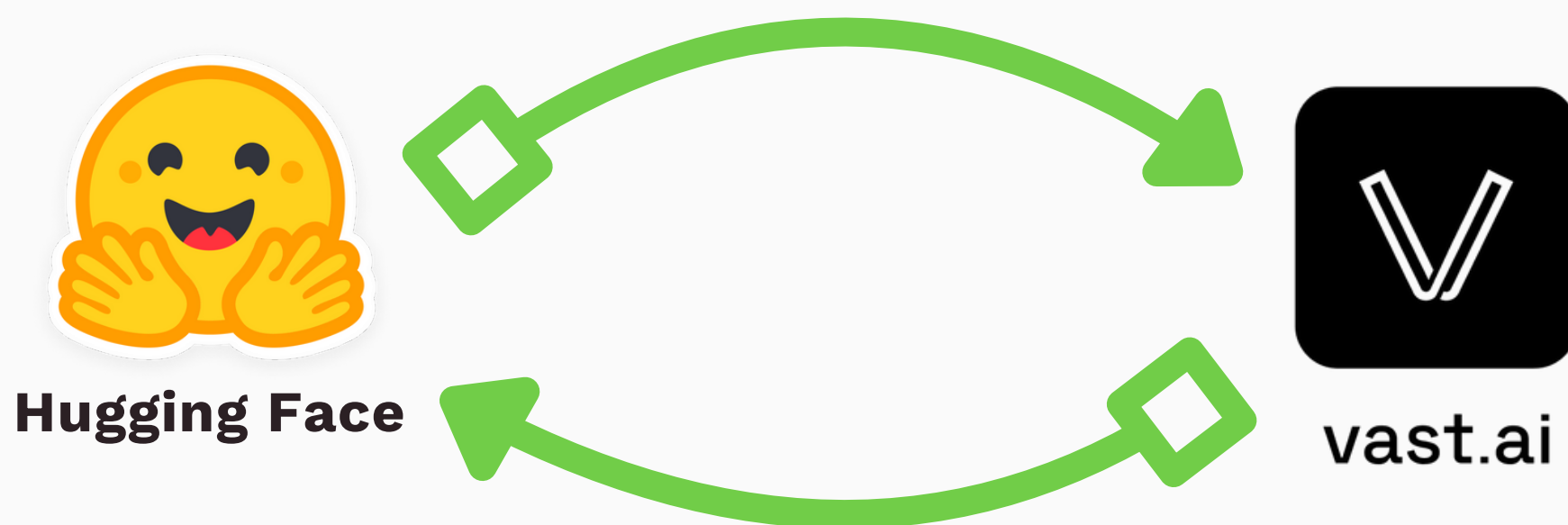


Specs and Investment

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Achieving this structured output required a focused, transparent training process:

- **Base Model:** [DeepSeek-R1-0528-Qwen3-8B](#)
- **Datasets:** [Indonesian Legal QA](#)
- **Fine-Tuning Method:** GRPO + Knowledge Distillation
(Recorded all generated respons and scores)
- **Training Specs:**
 1. **Steps:** 2,000 Training Steps with 16 generations per step
 2. **Epoch:** 1
 3. **Pipeline:** Cloud-to-cloud training (remote compute and storage)
 4. **Compute:** Nvidia RTX6000 Ada (via **vast.ai**)
 5. **Cost:** $\approx$ 50 USD



Structured Reasoning vs. Specificity

- **Findings:** show success in logic but highlight the limitations of small-scale training process

Positive Outcome (Success)	Limitation (R&D Finding)
Structured Reasoning: The model successfully adopted the 4-step legal reasoning framework (Context → Identification → Application → Conclusion).	Regulation Quoting: The model demonstrates limited accuracy in recalling or quoting the <i>specific article numbers</i> and <i>exact phrasing</i> of Indonesian regulations.
Legal Comprehension: Strong understanding and application of legal terms and concepts.	Scope Constraint: This limitation is a direct result of the small-scale, 2,000-step training scope , which prioritizes structural flow over deep, encyclopedic recall.

- **Impact:** The AI's output is highly valuable for preliminary analysis and structural guidance, but it must always be paired with verification by a legal expert.

- **Availability:**

- 1. **Model:** [Azzindani/Deepseek ID Legal Preview](#)
 - 2. **Last Checkpoint:** [Azzindani/DeepSeek_0528_8B_Legal_Ckpt](#)
 - 3. **Distilled Knowledge:** [Azzindani/Deepseek ID Legal Preview](#)



Structured AI and the Future Roadmap



- **Conclusion:** This research confirms that advanced policy-based fine-tuning (GRPO) can successfully instill structured, domain-specific reasoning—particularly in the Indonesian legal domain—into powerful LLMs, achieving our core goal of enhancing clarity. The project remains open-source to support transparency and ethical review.
- **Future Plan:**
 1. **RAG Integration:** Link the model to an up-to-date Indonesian legal database for accurate recall.
 2. **Reasoning Refinement:** Enhance step-by-step legal reasoning for greater clarity and consistency.
 3. **Reward Function Adjustment:** Tweak reward signals in GRPO to encourage better structured reasoning.
 4. **Quantization & Local Deployment:** Reduce model size to run locally on personal hardware.
- **Acknowledgements:**
 1. [Qwen team](#)
 2. [Deepseek team](#)
 3. [GRPO research paper](#)





This project is a proof-of-concept showing that transparent, structured reasoning in AI is possible across domains. It is an open R&D initiative.

**Interested in
AI fine-tuning or tech
that demands clear,
structured output?**

**Let's talk—your insights
matter for the next phase
of this open research.**



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